Akshi Robot

Client Demo Guide

12 Interactive Demo Scenarios

with Shortened Timers for Quick Presentations

Adaptive Robot-Assisted Education for Preschoolers

Powered by 5E Learning Model + Reinforcement Learning

Quick Start

Each scenario can be launched with a single command. Timer-heavy scenarios (motivation nudges, timeouts) use shortened timers so you can demo any flow in under 30 seconds.

Launch any scenario:

```
python demo.py <scenario_number>
```

Reset student data (SQLite only, Firebase untouched):

```
python reset_sqlite.py
```

Show all scenarios:

```
python demo.py 0
```

All Scenarios at a Glance

#	Scenario	Command	Timer	Key
1	Smart Student	python demo.py 1	Normal	Press 2
2	Calibration Game	python demo.py 2	Normal	Space
3	Wrong Answer	python demo.py 3	Normal	Press 1/3/4
4	Motivation Nudge	python demo.py 4	5s	Wait & watch
5	L1 Timeout	python demo.py 5	15s	Just wait
6	Rabbit Jump Break	python demo.py 6	10s	Watch + Enter
7	Break Return	python demo.py 7	Normal	Press Enter
8	Break Auto-Skip	python demo.py 8	15s	Don't press
9	L3 Auto-Skip	python demo.py 9	15s	Just wait
10	Teacher Intervention	python demo.py 10	Normal	Press C
11	Full Cascade	python demo.py 11	Fast	Wrong answers
12	RL Smart Start	python demo.py 12	Normal	Press 2

Detailed Scenario Descriptions

Scenario 1: Smart Student (Happy Path)

Student sees the Level 1 evaluation with 4 answer options. They press the correct answer (key 2 = Right Arrow) and the robot celebrates with voice and animation.

```
$ python demo.py 1
```

Keys:	`Press [2] for correct`	Timer:	`Normal`

Scenario 2: Calibration Game

The beautiful 'Magical Eyes' calibration mini-game. 3 acts: Owl (eyes open), Sleeping Bunny (eyes closed), Celebration (calibration complete). Press Space for auto-play.

```
$ python demo.py 2
```

Keys:	`[Space] auto / [1][2][3] manual`	Timer:	`Normal`

Scenario 3: Wrong Answer -> Kinesthetic

Student gives a wrong answer at L1. Robot gives encouragement speech, then escalates to the Kinesthetic activity screen where the teacher judges Pass [P] or Fail [F].

```
$ python demo.py 3
```

Keys:	`[1]/[3]/[4] wrong, then [P]/[F]`	Timer:	`Normal`

Scenario 4: 1-Minute Motivation Nudge

Student doesn't answer. After 5 seconds (shortened from 60s), the robot speaks a motivational message encouraging them to try. Timer continues running.

```
$ python demo.py 4
```

Keys:	`Wait 5s, then [2] to answer`	Timer:	`5s (was 60s)`

Scenario 5: L1 Timeout -> Kinesthetic

Student remains unresponsive. Motivation nudge at 5s, then full timeout at 15s (shortened from 180s). Robot automatically transitions to the Kinesthetic test.

```
$ python demo.py 5
```

Keys:	`Just wait and watch`	Timer:	`15s (was 180s)`

Scenario 6: Rabbit Jump Break (L2)

The full gamified bunny hop break! 3 acts: Story (robot tells child to hop), Activity (10s countdown with bunny hopping along a path), Return (press Enter to come back).

```
$ python demo.py 6
```

Keys:	`Watch, then [Enter]`	Timer:	`10s (was 45s)`

Scenario 7: Break Return -> Retry L2

Jumps directly to the Return phase of the break screen. Student presses Enter to 'come back' and the system retries the Level 2 evaluation.

```
$ python demo.py 7
```

Keys:	`Press [Enter]`	Timer:	`Normal`

Scenario 8: Break Auto-Skip

Student doesn't return after the break. The system counts down 15s (shortened from 60s), gives a verbal warning at 15s remaining, then automatically skips to the next student.

```
$ python demo.py 8
```

Keys:	`Don't press anything`	Timer:	`15s (was 60s)`

Scenario 9: L3 Auto-Skip

Level 3 student is completely unresponsive. Gentle motivation nudge at 5s, then at 15s (shortened from 180s) the robot says goodbye and skips to the next student.

```
$ python demo.py 9
```

Keys:	`Just wait and watch`	Timer:	`15s (was 180s)`

Scenario 10: Teacher Intervention

The final escalation. Teacher sees a gamified screen showing: the exact question, all answer options with images, the correct answer highlighted with a star, and the student's learning path.

```
$ python demo.py 10
```

Keys:	`Press [C] to continue`	Timer:	`Normal`

Scenario 11: Full Cascade (Interactive)

Walk through the entire escalation ladder: L1 -> Kinesthetic -> Engage -> L2 -> Explore -> L3 -> Explain -> L3 -> Elaborate -> L3 -> Teacher. Press wrong answers to progress.

```
$ python demo.py 11
```

Keys:	`Wrong keys + [Enter]`	Timer:	`Fast`

Scenario 12: RL Smart Start

Pre-seeds the database with student history, then shows how the robot intelligently skips to the optimal starting level. Demonstrates the Reinforcement Learning adaptation.

```
$ python demo.py 12
```

Keys:	`Press [2] for correct`	Timer:	`Normal`

Recommended Demo Order

For the best client impression, follow this order. Total presentation time: approximately 5 minutes.

Order	Scenario	Why Show This	Time
1	Calibration Game	Visual wow factor - stunning animated scenes	~30s
2	Smart Student	Happy path - shows the basic evaluation flow	~15s
3	Wrong Answer	Shows robot encouragement and escalation	~20s
4	Motivation Nudge	Shows the robot cares about the child	~15s
5	Rabbit Jump Break	Star feature - gamified physical activity break	~30s
6	Teacher Intervention	Shows question + correct answer for teacher	~15s
7	RL Smart Start	Shows intelligent adaptation to student history	~20s

Evaluation Timeout Behavior

Each evaluation level has different timeout behavior to handle unresponsive students appropriately:

Level	At 1 Minute	At 3 Minutes
Level 1	Gentle motivation nudge	Escalate to Kinesthetic test
Level 2	Strong motivation nudge	Rabbit Jump Break -> Retry L2
Level 3	Supportive gentle nudge	Auto-skip to next student

Learning Escalation Cascade

When a student answers incorrectly, the system follows this deterministic cascade. Each step provides more scaffolding before re-evaluating:

Step	Stage	Description
1	Evaluate L1	4 options, full question
2	Kinesthetic Activity	Physical learning (teacher judged)
3	Engage	Interactive video activity
4	Evaluate L2	2 options, simplified
5	Explore	Immersive storytelling
6	Evaluate L3	2 options, personalized
7	Explain	Direct concept explanation
8	Evaluate L3	Re-evaluation
9	Elaborate	Guided practice with hints
10	Evaluate L3	Final re-evaluation
11	Teacher Intervention	Human teacher takes over

Keyboard Controls Reference

Key	Action	Used In
1, 2, 3, 4	Select answer option	Evaluation screens
Enter	Confirm / Continue	5E screens, Break return
C	Teacher Continue	Teacher Intervention
P	Teacher Pass	Kinesthetic screen
F	Teacher Fail	Kinesthetic screen
B	Request Break	L2+ Evaluation
S	Skip question	L3 Evaluation
W	Wake robot	Idle screen
Space	Auto-play	Calibration game

Utility Commands

Reset Database (SQLite only)

```
python reset_sqlite.py
```

Clears all student metrics and session history from the local database. Firebase cloud data is NOT affected.

Run Full Application

```
python main.py
```

Launches the complete Akshi Robot interface with all screens, backend, and real timers.

Run Individual Demo

```
python demo.py <1-12>
```

Launches a specific scenario with shortened timers for quick presentations.